



SmartRepay AI

AI-Powered Loan Repayment Reconciliation & Posting System

Scope of Work

Client	Simplified Lending Bahamas
Product	SmartRepay AI — MRP Release
Duration	30 days
Audience	Development Team



1. Project Overview

SmartRepay AI is a focused operational tool that automates loan repayment reconciliation and posting for Simplified Lending Bahamas. It ingests daily bank statements, matches transactions to borrowers, flags anything it cannot confidently resolve, and prepares clean repayment data for LoanDisk and QuickBooks.

Primary outcomes:

- Cut manual processing time across the reconciliation workflow.
- Eliminate unapplied payments caused by naming and reference mismatches.
- Replace ad-hoc WhatsApp handoffs with a tracked, auditable workflow.
- Provide a single operational view for collections, mid-office, and accounting.

2. Problem Statement

Today, repayment processing is manual, repetitive, and dependent on institutional knowledge held by specific people. Bank transactions arrive with inconsistent borrower names, business names instead of personal names, or missing references. Staff identify payments by memory, confirm them over WhatsApp, and hand them off through multiple parties before they reach LoanDisk and QuickBooks. The result is unapplied payments, slow reconciliation, and high operational overhead.

3. MRP Scope

The first release is a Minimum Remarkable Product: one workflow, fully functional end-to-end, delivered inside a controlled environment. Core loop:



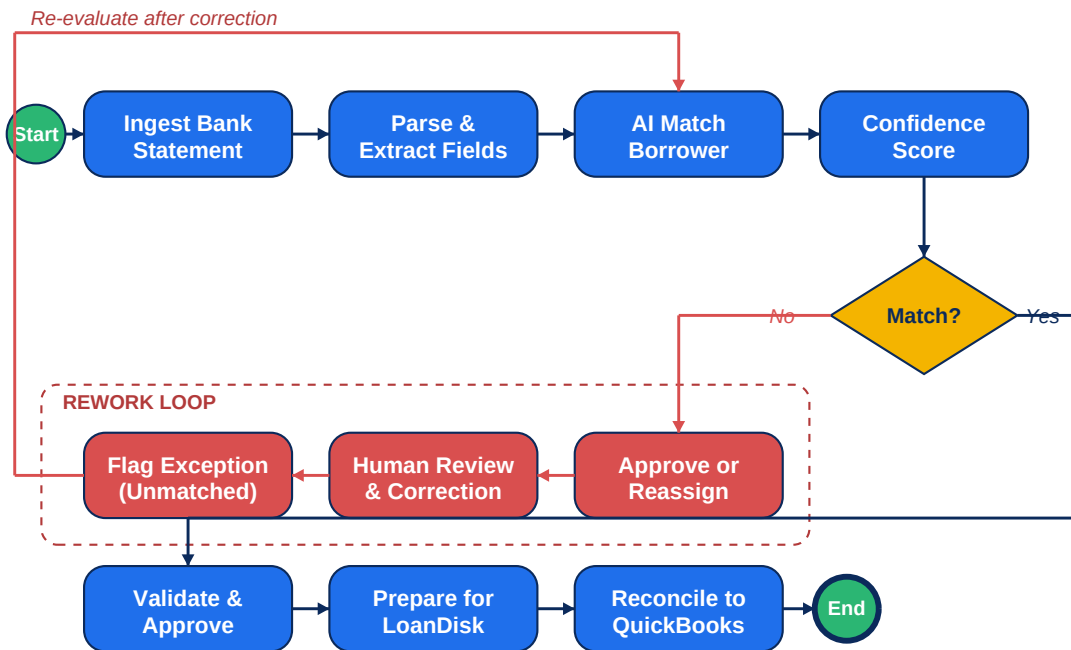
4. Process Flow

The diagram below is the target end-to-end flow using standard BPMN notation: circle for start/end, rounded rectangle for task, diamond for decision. Red tasks are the exception path.



SmartRepay AI — End-to-End Process Flow (BPMN)

● Start ■ Task ◆ Decision ■ End



5. Current vs. Target Workflow

Stage	Current	Target with SmartRepay AI
Intake	Manual download and visual scan of bank statements.	Automated ingestion and field extraction.
Identification	Staff recall, employer references, WhatsApp.	AI matching against borrower and employer history.
Exceptions	Ad-hoc clarification across people.	Flagged queue with reviewer assignment.
Posting	Manual entry into LoanDisk.	Pre-validated payload prepared for posting.
Reconciliation	Spreadsheet tracking, multiple handoffs.	Tracked workflow with audit trail.

6. System Architecture

Ingestion Service	Parses bank statements; extracts name, description, amount, employer reference, timestamp.
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Matching Engine	AI-assisted matching against borrowers, employers, and historical patterns; produces a confidence score.
Exception Engine	Detects unmatched, duplicate, partial, or suspicious entries.
Workflow Engine	Routes items through validate → approve → prepare-to-post.
Review UI	Queues, filters, side-by-side compare, one-click confirm/correct.
Audit & Logging	Immutable record of matches, edits, approvals, and user actions.
Notifications	Status updates, exception alerts, daily reconciliation summary.
Secure Environment	Role-based access; financial data never leaves the controlled environment.

7. Functional Requirements

7.1 Statement Ingestion

- Accept PDF and CSV bank statement formats used today.
- Normalize records to a common schema (date, payer, description, amount, reference).
- Idempotent ingestion: re-uploading the same statement does not duplicate transactions.

7.2 Matching Engine

- Match by borrower name, alias, employer reference, and prior payment fingerprint.
- Return a confidence score with each suggested match.
- Above a configurable threshold → auto-approve queue; below → exception queue.

7.3 Exception Handling

- Flag unmatched, duplicate, partial, and suspicious records.
- Allow reviewer to confirm, reassign, split across loans, or reject.
- Capture the reason for every manual override.

7.4 Exception Aging & SLA Tracking

- Timestamp every exception at creation and track time-in-queue until resolved.
- Configurable SLA per exception type (e.g., 4h, 24h, 72h) with aging buckets: On Track, At Risk, Breached.
- Automatic escalation alerts to the assigned reviewer and supervisor when SLA is at risk or breached.
- Aging dashboard with counts by bucket, reviewer, and exception type for management reporting and operational accountability.

7.5 Validation & Posting Prep

- Reviewer screen with bank record on the left and proposed borrower/loan on the right.
- Approved items produce a LoanDisk-ready payload (export or API, per integration decision).



- Reconciliation view aligns posted items to bank statement totals.

7.6 Notifications & Tracking

- Daily summary: processed, posted, pending, exceptions.
- Per-exception alert to the assigned reviewer.

7.7 Audit & Logging

- Every action (match, edit, approve, reject) is logged with user, timestamp, and prior value.
- Logs are queryable and exportable.

8. Non-Functional Requirements

Area	Requirement
Security	Role-based access; no financial data sent to public AI services.
Reliability	Daily reconciliation completes within standard business hours.
Performance	Process a typical day's statement volume in under 5 minutes.
Auditability	100% of state changes captured in the audit log.
Usability	Reviewer can clear a typical exception in under 30 seconds.

9. Roles & Responsibilities

Role	Responsibility
Collections Team	First-line review; clarify exceptions surfaced by the system.
Mid-Office / Operations	Validate postings and confirm reconciliation accuracy.
Accounting Team	Final reconciliation and QuickBooks posting.
System Owner	Workflow oversight; threshold and rule configuration.
Development Team	Build, deploy, and support the MRP per this SOW.

10. Delivery Timeline (30 Days)

Week	Focus	Deliverables
1	Discovery & Data Alignment	Workflow map, statement format spec, matching rules, infrastructure plan.
2	Core Development	Ingestion service, matching engine, exception detection.



3	Workflow & UI	Reviewer UI, notifications, audit logging.
4	Pilot, Tune, Release	User testing, threshold tuning, pilot deployment, handover.

11. Success Metrics

Baseline measurements are captured before go-live: daily processing time, manual reconciliation time, unapplied-payment count, correction count, manual touchpoints, and daily repayment volume. The release is considered successful when:

Executive KPI

- Repayment posting turnaround time reduced by 50% or more (statement received → posted to LoanDisk / reconciled to QuickBooks), with a measurable reduction in reconciliation backlog.

Operational KPIs

- Manual processing effort reduced by 25% or more.
- Repayment matching accuracy $\geq 90\%$.
- Measurable drop in unapplied payments and reconciliation time.
- Fewer manual handoffs across the workflow.
- $\geq 95\%$ of exceptions resolved within SLA.
- Positive reviewer feedback on the new workflow.

12. Risks & Mitigations

Risk	Mitigation
Incorrect repayment match	Confidence threshold plus human approval before posting.
Unmatched payments	Exception queue with reviewer assignment.
Inconsistent statement formats	Schema-mapping layer in the ingestion service.
User error	Validation checkpoints and audit log.
Dependency on individuals	Documented workflow inside the tool.

13. Governance, Security & Data Protection

- AI assists; humans approve. No autonomous posting in the MRP.
- Role-based access; audit log captures every state change.
- Financial data stays within the controlled environment — no third-party AI processing of raw data.
- All workflow approvals are traceable to a named user.



14. Boundaries & Out of Scope

In Scope (MRP)	Out of Scope (MRP)
AI-assisted matching, exception handling, reconciliation support, LoanDisk-ready payloads, and audit trail.	Full accounting automation, direct write-back to QuickBooks, WhatsApp replacement, and ERP replacement.

15. Versioning Roadmap

V1 — MRP	V2	V3
AI-assisted matching, exception handling, and reconciliation support.	Stronger matching intelligence, workflow tuning, and expanded reporting.	Additional integrations and deeper end-to-end automation.

16. Next Steps

- Confirm bank statement sources and access method.
- Confirm LoanDisk integration approach (export vs API).
- Lock the matching rules and confidence thresholds for V1.
- Schedule pilot users and the baseline measurement window.
- Kick off Week 1 of the timeline.

SmartRepay AI streamlines the operational work that supports the core financial processes of Simplified Lending Bahamas — designed to assist, not replace, the people who own the workflow.